

Stat 3113 – Exam 3 – Topics & Key Words

Lesson 26: One-Way ANOVA Part 1

- Factor, Levels, Response Variable, Number of Replicates
- Hypothesis Test to Compare Three or More Means
- Assumptions (Normality, Equal Variance, Independence)
- ANOVA Table (Source, df, SS, MS, F test statistics, P-value)
- Significance Level
- Decision Rule (Reject/Fail to Reject)
- Multiple Comparison Tests (Individual/Familywise Confidence Levels)
- Tukey-Kramer HSD Post Anova Test (Connecting Letters Report, Ordered Differences Report)
- Assumption Checking (Normal Probability, Residual vs Factor, etc.)

Lesson 27: One-Way ANOVA Part 2

- Practice Examples

Lesson 28: Two-Way ANOVA

- Treatment Combinations, Row Factor, Column Factor, Complete Design, Full Factorial Design, Balanced Experiment, Fixed/Random Effects, etc.
- Main Effects, Interaction Effects, Interaction Plots, Hypotheses Testing Procedure, Two-Way ANOVA Table, Checking Assumptions, Application Examples, etc.

Lesson 29: 2^3 Factorial Experiment

- Main Effects, Two-Way Interaction, Three-Way Interaction, Treatment Combination, ANOVA Table, Hypotheses Testing Procedure, Checking Assumptions

Lesson 30: ANOVA Examples

Lesson 31: Additional ANOVA examples covering 2^3 and 3^2 factorial experiments.

Lesson 32: Bivariate Analysis – Scatterplots

- Bivariate Analysis, Quantitative Variable, Response Variable, Explanatory Variable, Scatterplots Interpretation (Outliers, Clusters, Association, Functional Form)

Lesson 33: Bivariate Analysis – Correlation

- Pearson’s correlation (Definition and Properties)

Lesson 34: Regression Analysis

- Least-Squares Regression, Squared Error, Simple Linear Regression, Fitted Line, Line Plot, Intercept b_0 , Slope b_1 , Regression Equation (pay attention to the hat), Interpretation of the coefficients.
- Testing for the Significance of the Slope Parameter, ANOVA, P-value from Output
- Coefficient of Determination r^2 and its interpretation
- Checking Assumption (Normal Probability Plot and Residual Plot)
- Cautions (Extrapolation, Association/Causation, Gaps in the Data)

Lesson 35: Regression Examples